

THE FREE ENERGY OF CHOICE

Toward a Unified Architecture for Decision Intelligence in Biological and Artificial Systems

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*"The purpose of a brain—and by extension any intelligent system—is
not to see the world as it is, but to predict what will happen next and
act so that bad things don't."*

— Karl Friston, paraphrased

Abstract

Five independent research programmes—world models (LeCun, Hafner), neuro-symbolic reasoning (Garcez, Riegel), causal representation learning (Schölkopf, Pearl), active inference (Friston), and drift-diffusion decision theory (Ratcliff, Kraglich)—have converged on architecturally equivalent prescriptions for intelligent decision-making without acknowledging the convergence. This paper demonstrates that the convergence is not metaphorical but mathematical: all five frameworks minimise the same variational functional under different parameterisations. We formalise this claim as the **Variational Decision Hypothesis** and derive a five-layer architecture—the Neuro-Symbolic Decision Frame (NSDM-F)—that implements the unified principle. We then present the paper's primary original contribution: an extension of Heins, Friston, and Couzin's (2024) surprise-minimisation formalism for biological collective behaviour to economic and consumer-market dynamics, showing that herding, brand loyalty, market panics, and cultural taste formation emerge naturally from agents minimising variational free energy

over choice-relevant latent states. Finally, we argue that this architecture’s information-theoretic structure inherently favours small, domain-specialised models over monolithic scaling—a prediction with direct implications for AI deployment in infrastructure-constrained markets that frontier labs systematically ignore. We propose seven falsifiable predictions and an open experimental protocol.

1. The Problem of Five Bridges

In 1735, Euler proved that no walk could cross all seven bridges of Königsberg exactly once. The problem was not about bridges. It was about the structure of connectivity itself—the birth of graph theory. We face an analogous moment in artificial intelligence. Five major research programmes have independently built bridges toward the same island—a complete theory of decision under uncertainty—without recognising that they are building toward the same destination.

The programmes are: (1) **Predictive world models**, from LeCun’s Joint Embedding Predictive Architecture (JEPA) through Hafner’s DreamerV3, which propose that intelligence is fundamentally about predicting the consequences of actions in a learned latent space. (2) **Neuro-symbolic integration**, from Garcez through IBM’s Logical Neural Networks (Riegel et al., 2020) to DeepMind’s AlphaGeometry, which demonstrates that compositional reasoning requires symbolic structure that pure neural networks cannot reliably learn. (3) **Causal representation learning**, Schölkopf’s programme to build systems that distinguish correlation from causation by learning representations invariant under intervention. (4) **Active inference**, Friston’s free energy principle, which casts perception, action, and learning as a single process of variational free energy minimisation. (5) **Drift-diffusion decision theory**, the dominant computational model of biological choice, which describes decisions as noisy evidence accumulation toward commitment boundaries.

Each programme has produced extraordinary results in isolation. V-JEPA 2 (Assran et al., 2025) achieves zero-shot robot control from video prediction alone. AlphaGeometry 2 (Chervonyi et al., 2025) solves 84% of International Mathematical Olympiad geometry problems by coupling a neural language model with a symbolic deduction engine. Heins, Friston, and Couzin (2024) demonstrate that milling, cohesion, and directed collective motion emerge from active inference without any explicitly programmed behavioural rules. Amortised neural drift-diffusion estimators (Fengler et al., 2022; Zammit-Mangion et al., 2025) have made Bayesian inference over human decision processes computationally tractable for the first time.

Yet these programmes cite each other rarely and disagree about what intelligence *is*. LeCun frames intelligence as world-model-conditioned action selection. Friston frames it as surprise minimisation. Pearl frames it as causal intervention. Ratcliff frames it as evidence accumulation. Garcez frames it as neural-symbolic integration. They cannot all be right about what intelligence is—unless they are all describing different faces of the same mathematical object.

Thesis. They are. This paper demonstrates that all five frameworks minimise the same variational functional. The disagreements are about parameterisation, not principle. And the unified principle has implications that none of the individual programmes has drawn—in particular, for how intelligent systems should be designed to make decisions in the real world, under constraints that frontier AI labs are structurally incentivised to ignore.

2. The Variational Decision Hypothesis

We state the central claim precisely, then show how each programme instantiates it.

2.1 Statement

Definition (Variational Decision Hypothesis). Any system that (a) maintains an internal model of its environment, (b) selects actions that change its relationship to that environment, and (c) persists over time, can be described as minimising a functional of the form:

$$F[q] = E_q[\ln q(\theta) - \ln p(o, \theta)] = D_{KL}[q(\theta) \parallel p(\theta | o)] - \ln p(o)$$

where o denotes observations, θ denotes the parameters of the agent's generative model (hidden states, causes, model parameters), $q(\theta)$ is an approximate posterior (the agent's beliefs), and $p(o, \theta)$ is the generative model (how the agent believes observations are caused). Because $D_{KL} \geq 0$, free energy F is an upper bound on surprise $-\ln p(o)$. Minimising F with respect to q (perception) makes beliefs accurate. Minimising F with respect to actions (active inference) makes observations conform to expectations. Minimising F with respect to model parameters (learning) makes the generative model better.

The hypothesis is that this functional is not merely a useful objective for engineering AI systems. It is the *only* objective any persistent decision-making system can be coherently described as optimising, in the same sense that the second law of thermodynamics is not a design choice but a constraint on physical systems.

2.2 The Five Instantiations

World models (LeCun/Hafner). The JEPA objective predicts target representations s_y from context representations s_x by minimising $D(s_y, \text{Pred}(s_x))$ in a learned latent space. DreamerV3's RSSM training objective is the variational ELBO: $L = -E_q[\ln p(o_t | s_t)] + \beta \cdot D_{KL}[q(z_t | h_t, o_t) \parallel p(z_t | h_t)]$. This is variational free energy under a sequential generative model. LeCun's JEPA differs from Friston's formulation only in that it avoids explicit probability distributions, using contrastive or distillation losses instead—but VICReg and Barlow Twins regularisers are information-theoretic surrogates for the KL term. The JEPA objective and the free-energy functional are related by: $F_{JEPA} = E[\|s_y - \text{Pred}(s_x)\|^2] + R(s)$ where $R(s)$ is a representation collapse regulariser that serves the same role as the KL divergence—preventing the approximate posterior from trivialising.

Neuro-symbolic reasoning (Garcez/Riegel/AlphaGeometry). IBM's LNN maintains truth-value bounds $[l, u] \in [0, 1]^2$ for each proposition, with inference as message passing that tightens these bounds under logical constraints. This is constraint satisfaction over a factored belief distribution—formally, variational inference on a factor graph where some factors are deterministic (logical rules) and others are learned (neural weights). AlphaGeometry's two-system architecture—neural language model generating hypotheses, symbolic deduction engine verifying them—implements a proposal-verification loop that is precisely the structure of variational inference: the neural model proposes $q(\theta)$, the symbolic engine evaluates $p(o, \theta)$, and the search algorithm minimises the gap.

Causal representation learning (Schölkopf/Pearl). Schölkopf et al. (2021) argue that robust generalisation requires representations that are invariant under intervention—i.e., representations that factor according to the causal graph of the data-generating process. The minimum description length (MDL) principle that selects causal over anticausal

models is itself a free-energy minimisation: $MDL = -\ln p(D / M) + L(M)$ where $L(M)$ is the model complexity (analogous to the KL term). Pearl's do-calculus provides the algebra for computing interventional distributions within the generative model—it tells you what $p(o / do(a))$ looks like when you surgically modify the causal graph, which is exactly what active inference does when selecting actions.

Active inference (Friston). Active inference is the explicit, fully general formulation of the variational decision hypothesis. What we add is not the mathematics—Friston's group has done that work—but the demonstration that the other four programmes, developed largely independently, converge on the same functional. Active inference's unique contribution is the *expected free energy* $G(\pi) = E_{-q}[F(o, \theta / \pi)]$ over policies π , which decomposes into an epistemic (information-seeking) term and a pragmatic (goal-seeking) term: $G = -\text{Information Gain} - \text{Expected Value}$. This decomposition has no analogue in the other four frameworks as currently stated—it is the missing piece that turns them from learning systems into acting systems.

Drift-diffusion models (Ratcliff/Krajibich). The DDM describes evidence accumulation as a stochastic differential equation $dx = \nu \cdot dt + \sigma \cdot dW$ with drift rate ν (signal strength) and diffusion coefficient σ (noise). A decision is made when accumulated evidence x hits a boundary. The connection to free energy: the log-likelihood ratio of two hypotheses under sequential sampling is a drift-diffusion process, and the optimal boundary is set by the expected free energy of continuing to sample versus committing. Friston (2012) showed this explicitly: the DDM is a special case of active inference where the only available actions are 'sample more evidence' and 'commit to a decision.' The drift rate $\nu = -\partial F / \partial \mu$ is the gradient of free energy with respect to beliefs.

2.3 The Correspondence Table

Table 1 summarises the five-way correspondence:

Component	World Models	NeSy	Causal RL	Active Inference	DDM
Objective	Latent prediction error	Logical consistency loss	Interventional invariance (MDL)	Variational free energy F	Log-likelihood ratio

Generative model $p(o, \theta)$	Decoder + dynamics model	Ontology + rules	Structural causal model	Generative model	Signal + noise model
Approximate posterior $q(\theta)$	Encoder output	Belief bounds $[l, u]$	Learned representation	Recognition density	Accumulated evidence x
KL / complexity term	VICReg / Barlow Twins	Rule weight priors	Model description length $L(M)$	$D_{KL}[q p]$	Boundary setting cost
Action selection	Actor-critic in latent space	Constrained optimisation	do-calculus intervention	Expected free energy $G(\pi)$	Commit at boundary
Unique contribution	Sample-efficient planning	Compositional OOD reasoning	Causal invariance	Epistemic-pragmatic decomposition	Neural decision dynamics

Table 1. The five-way correspondence. Each row maps a structural component of the variational decision hypothesis to its instantiation in each research programme.

3. NSDM-F: The Five-Layer Architecture

The Variational Decision Hypothesis implies an architecture. If all five programmes are minimising the same functional, then a complete decision-intelligence system must implement all five simultaneously—not as a pipeline where one feeds the next, but as a recurrent loop where each layer constrains and is constrained by the others, all driven by a single variational objective. We call this the **Neuro-Symbolic Decision Frame (NSDM-F)**.

Layer 1: The Latent World Model

A JEPA-class latent prediction engine. Takes multimodal observations (visual stimuli, text, contextual metadata) and learns compressed representations in which future states can be predicted without reconstructing raw observations. The world model maintains both deterministic (predictable dynamics) and stochastic (uncertainty, multimodality) state components, following the RSSM architecture proven in DreamerV3. Training objective: the variational ELBO with a symlog observation transform for scale invariance across diverse domains.

Formal specification: $s_t = [h_t, z_t]$ where $h_t = f_{det}(h_{t-1}, z_{t-1}, a_{t-1})$ is deterministic recurrence and $z_t \sim q(z_t | h_t, o_t)$ is stochastic state. Loss: $L_1 = -E_q[\ln p(o_t | s_t)] + \beta \cdot D_{KL}[q(z_t | h_t, o_t) \| p(z_t | h_t)]$.

Layer 2: The Causal-Symbolic Reasoning Engine

A neuro-symbolic layer combining two sub-components: (a) an LNN-class constraint system encoding domain knowledge, behavioural principles, and ethical rules as differentiable logical formulae with learnable weights and truth-value bounds; and (b) a structural causal model specifying the causal graph over decision-relevant variables, enabling interventional reasoning via the do-calculus. The causal graph provides the *structure* of the generative model; the LNN provides the *constraints* under which inference proceeds.

Formal specification: Causal graph $G = (V, E)$ with structural equations $v_i = f_i(pa(v_i), u_i)$. LNN constraint set $C = \{c_j : w_j \cdot \varphi_j(\theta) \geq \tau_j\}$ where φ_j are differentiable logical formulae

and w_j are learnable confidence weights (fixed at 1.0 for ethical hard constraints). Loss: $L_2 = L_{causal} + \lambda_1 \cdot L_{LNN} + \lambda_2 \cdot L_{constraint}$.

Layer 3: The Decision Accumulator

A neural drift-diffusion model that parameterises evidence accumulation as a learned function of the latent state from Layer 1, constrained by the causal-symbolic reasoning from Layer 2. The drift rate $v(s_t; \theta)$ becomes a differentiable function of the world model's latent representation, and the decision boundaries $a(c; \psi)$ are conditioned on context (audience, culture, constraints). This layer provides the temporal dynamics of decision-making—it models *how long* and *how confidently* an agent commits to a choice, not just what it chooses. Amortised Bayesian inference via Likelihood Approximation Networks enables real-time parameter estimation.

Layer 4: The Active Inference Policy

An expected-free-energy policy layer that selects actions to minimise $G(\pi) = E_q[-\ln p(o | \pi) + D_{KL}[q(\theta | o, \pi) \parallel q(\theta | \pi)]]$. The first term is pragmatic value (do outcomes match preferences?). The second term is epistemic value (does the action reduce uncertainty about hidden states?). This decomposition is the key architectural innovation over standard reinforcement learning: the agent *naturally* explores to reduce uncertainty and exploits to achieve goals, without needing separate exploration bonuses or curiosity rewards. Action selection is constrained by Layer 2's ethical hard constraints: actions that violate $LNN(a, s) < \tau$ are removed from the policy space before evaluation.

Layer 5: The Information-Theoretic Objective

The unifying loss function that binds Layers 1–4 into a single variational objective:

$$L_{total} = L_1(world\ model) + L_2(causal-symbolic) + L_3(decision\ dynamics) + G(policy) + \alpha \cdot I(S; A | O)$$

where $I(S; A | O)$ is the conditional mutual information between latent states and actions given observations—an information-bottleneck regulariser that prevents the system from encoding more complexity in its actions than is justified by the information content of the environment. This term connects to rate-distortion theory: the agent should use the

minimum-complexity policy that achieves its goals. The total loss is, under appropriate conditions, equivalent to the variational free energy of the entire agent-environment system.

4. From Flocking to Markets: Surprise Minimisation in Economic Collectives

This section contains the paper’s primary original contribution.

4.1 The Heins-Friston-Couzin Result

Heins, Friston, and Couzin (PNAS, 2024) demonstrated that classic collective behaviours—cohesion, milling, directed motion—emerge when individual agents minimise variational free energy over sensory states encoding the positions of neighbours, without any explicitly programmed social forces, attraction zones, or alignment rules. Each agent maintains a generative model of its expected sensory input (average nearest-neighbour distances across visual sectors), updates beliefs via precision-weighted prediction errors, and acts to minimise the discrepancy between predicted and observed neighbour configurations. The milling regime—agents orbiting a common centre—is stable across a far wider parameter range than in classical self-propelled particle models, and imprecise sensory inference is not merely tolerable but actively beneficial for collective cohesion.

The mathematical core: each agent’s hidden states are $\mu = \{\mu_1, \dots, \mu_L\}$ representing expected average neighbour distances per visual sector. Belief dynamics follow $d\mu/dt = -\nabla_{\mu} F$ where F is the variational free energy as a function of precision-weighted prediction errors between expected and observed sector distances. Action (heading direction change) follows $da/dt = -\nabla_a F$. No communication, no global information, no alignment rules. Flocking emerges from surprise minimisation alone.

4.2 The Extension: Consumer Markets as Active Inference Collectives

We propose that the Heins-Friston-Couzin formalism applies directly to consumer markets with the following structural mapping:

Biological Flocking	Consumer Market	Mathematical Object
Individual bird/fish	Individual consumer	Active inference agent with generative model
Position in physical space	Position in preference space	Latent state s_t
Neighbours in visual sectors	Visible choices of social peers	Observations o_t
Expected neighbour distance	Expected peer behaviour (brand	Prior beliefs $p(o \theta)$

	norms, social proof)	
Heading direction	Purchase/choice action	Policy π
Cohesion (staying near flock)	Brand loyalty / herd behaviour	Low variational free energy at steady state
Milling (circular orbiting)	Consideration-set cycling	Oscillatory attractor in belief dynamics
Directed motion (migration)	Market shift / trend adoption	Symmetry-breaking via precision modulation
Predator avoidance	Loss aversion / risk aversion	High prior precision on negative-valence observations
Imprecise sensing aids cohesion	Uncertainty drives social conformity	Low sensory precision increases social-prior weight

Table 2. Structural mapping from biological flocking (Heins-Friston-Couzin, 2024) to consumer market dynamics under the active inference formalism.

4.3 Emergent Market Phenomena from Surprise Minimisation

The mapping in Table 2 is not a metaphor. It is a formal claim that consumer markets exhibit the same dynamical regimes as biological collectives, for the same mathematical reasons. We derive four specific emergent phenomena:

Phenomenon 1: Brand loyalty as cohesion. When consumers maintain generative models predicting that ‘people like me buy brand X,’ deviating from brand X generates prediction error (surprise). Minimising surprise drives repeat purchase—not because of utility maximisation, but because switching brands violates the agent’s model of its social niche. This explains brand loyalty’s persistence even when objectively superior alternatives exist, and predicts that loyalty is stronger in collectivist cultures (higher precision on social-sector observations) than in individualist cultures.

Phenomenon 2: Market panics as flock predator-avoidance. In the Heins-Friston-Couzin model, high prior precision on threat-sector observations produces rapid, coordinated escape behaviour. In markets, loss aversion (Kahneman and Tversky, 1979) functions as high prior precision on negative-valence signals. When multiple agents simultaneously increase their precision on loss-related observations, the collective undergoes a phase transition analogous to predator-triggered flock dispersal—a market panic. The model predicts that panics are *easier* to trigger in markets with higher social

visibility of losses (transparent order books, social media) because visibility increases the effective precision on social-sector observations.

Phenomenon 3: Trend adoption as directed motion. In biological collectives, directed migration emerges when a small proportion of informed individuals modulate their prior precision, creating an asymmetry that the flock follows. In consumer markets, influencers and early adopters function as informed individuals whose high-confidence choices create asymmetric prediction errors for their social neighbours, triggering cascading belief updates. The model predicts a critical threshold of informed agents (approximately 5–10% of the social network, consistent with both Couzin et al.’s biological findings and the Rogers diffusion curve’s innovator percentage) required to trigger market-wide adoption.

Phenomenon 4: Cultural taste formation as milling. The milling regime—agents orbiting a common centre without settling on a fixed direction—corresponds to the phenomenon of cultural taste: shared preferences that persist and circulate within a community without converging to a single point. Fashion cycles, genre preferences, and aesthetic norms are milling behaviours in preference space. The Heins-Friston-Couzin result that milling is stable across a wider parameter range than in classical models predicts that cultural tastes are more robust than rational-choice models would suggest—precisely what is observed empirically.

4.4 The Cross-Cultural Precision Hypothesis

The most testable prediction emerging from this framework concerns cross-cultural variation. In the active inference formalism, the key parameter governing collective behaviour is *sensory precision*—the confidence an agent places on its own observations versus its prior expectations. Heins et al. (2024) show that low sensory precision (agents that don’t fully trust their own senses) produces *stronger* collective cohesion, because the prior (social expectation) dominates.

We hypothesise that Hofstede’s cultural dimensions—specifically the individualism-collectivism axis—map directly onto the sensory precision parameter in consumer active inference. Collectivist cultures (South Africa’s Nguni cultural cluster, East Asian markets)

effectively operate with lower individual sensory precision and higher social-prior precision, producing stronger brand cohesion, greater resistance to individual-targeted marketing, and faster collective trend adoption when the critical threshold is reached. Individualist cultures (US, Northern Europe) operate with higher sensory precision, producing weaker cohesion but greater responsiveness to individual-level persuasion. This is a falsifiable, quantitative prediction that connects cultural psychology to dynamical systems theory through the free energy principle.

5. The Efficiency Thesis: Why Architecture Beats Scale

Leopold Aschenbrenner’s Situational Awareness (2024) makes the case that raw computational scale—measured in floating-point operations per training run—is the primary driver of AI capability progress, with algorithmic improvements providing a secondary but significant multiplier. His hedge fund, now managing approximately \$5.5 billion, has bet on this thesis by investing in power infrastructure and data-centre capacity while exiting semiconductor positions.

We argue that Aschenbrenner’s infrastructure thesis is correct—power and cooling are genuine bottlenecks—but his architectural thesis is incomplete. The Variational Decision Hypothesis implies a different scaling law for *decision* systems than for *prediction* systems. Here is why.

The information-theoretic objective in Layer 5 of NSDM-F includes an information-bottleneck term $I(S; A / O)$ that penalises policy complexity beyond what the environment’s information content justifies. In a domain with low intrinsic dimensionality—consumer decisions over a fixed product category, financial risk assessment within a regulated framework, clinical diagnosis within a bounded symptom space—the optimal agent has *low policy complexity*. It does not need a 400-billion-parameter model because the decision space does not contain 400 billion parameters’ worth of relevant structure.

This is not a theoretical abstraction. Microsoft’s Phi-4 (14B parameters) matches models 5× its size on mathematical reasoning. Muckatira et al. (2024) show that domain-pretrained models below 165M parameters exhibit emergent capabilities exceeding larger general models in their speciality. The MIT Technology Review declared small language models one of the ten breakthrough technologies of 2025. The scaling law for domain-specific decision intelligence is not $performance \propto compute^{0.5}$ (the Kaplan-Hoffmann power law for general language modelling) but closer to $performance \propto (domain\ data\ quality) \times (architectural\ fit)$ with diminishing returns from raw scale.

A striking recent demonstration makes the point concrete. Hegedus (2026) trains a three-layer feedforward network to add ten-digit numbers at 93.5% accuracy using only 1,000

examples—by encoding the correct inductive bias (framing the carry operation as a directed-percolation phase transition) rather than by scaling. A task assumed to need recurrent architectures or vast data is solved by a tiny network with the right structure. This is the efficiency thesis in its purest form: architectural fit, not parameter count, determines competence on a structured task. It is independent evidence, on an unrelated problem, that the binding constraint is structure rather than scale.

The implication for emerging markets is structural, not incidental. South Africa’s power grid operates under chronic load-shedding constraints. The Stanford HAI AI Index 2025 notes that AI access remains limited in many African countries due to basic electricity infrastructure gaps. An architecture that *requires* trillion-parameter models is an architecture that *excludes* 1.4 billion Africans from the decision-intelligence frontier. An architecture grounded in the Variational Decision Hypothesis naturally favours small, domain-specialised models with causal structure and symbolic constraints—models that can run on constrained infrastructure while outperforming monolithic systems on bounded decision tasks.

Furthermore, the regulatory landscape reinforces this architectural advantage. South Africa’s POPIA Section 71 requires that automated decisions affecting individuals be explicable and contestable. The EU AI Act mandates transparency for high-risk AI systems. A pure neural system with hundreds of billions of parameters cannot satisfy these requirements. An NSDM-F system with an explicit causal graph, symbolic constraint traces, and interpretable decision boundaries can. The regulatory environment selects for the architecture the mathematics already recommends.

6. Seven Falsifiable Predictions

A theory that cannot be wrong is not a theory. We commit to seven predictions, each testable with existing or near-term methods:

P1: Brand loyalty correlates with collectivism score. Brand-switching rates will be significantly lower in high-collectivism markets (SA Nguni cluster, East Asia) than in high-individualism markets (US, Northern Europe), even controlling for income and product

availability, because collectivism maps to higher social-prior precision in the active inference formalism. *Test: Cross-national consumer panel data (Euromonitor, Nielsen) + Hofstede scores. Regression with cultural-dimension controls.*

P2: Market panics are precision-modulated phase transitions. Markets with higher social visibility of trading activity (transparent order books, social media sentiment feeds) will exhibit lower critical thresholds for panic onset and faster panic propagation, because visibility increases effective sensory precision on social-loss signals. *Test: Agent-based simulation calibrated to order-book transparency data across markets. Empirical validation against VIX/JSE SAVI flash-crash event studies.*

P3: Trend adoption follows the 5–10% informed-agent threshold. Consumer trend adoption will show a critical transition at 5–10% social-network penetration by early adopters, consistent with the Heins-Friston-Couzin finding for directed motion in biological collectives, and predictable from the precision parameters of the social generative model. *Test: Network analysis of product-adoption cascades on social-commerce platforms. Agent-based simulation with calibrated precision parameters.*

P4: SLMs with causal structure outperform LLMs on bounded decisions. A Phi-4-class (14B) model with explicit causal graph and symbolic constraints will outperform GPT-4-class models on consumer-decision prediction tasks where the causal graph is known, as measured by AUC-ROC on purchase-intent prediction and calibration on counterfactual creative-change recommendations. *Test: A/B test: NSDM-F (14B backbone) vs GPT-4V zero-shot vs CLIP+linear baseline on the Buyology Consumer Decision Benchmark (to be constructed).*

P5: Cross-cultural robustness requires causal, not just correlational, modelling. Models trained on Western consumer data will degrade by >15% AUC-ROC on South African consumer data (LSM 4–7 segments) for correlational architectures, but <5% for causal architectures with explicit cultural-dimension moderators in the causal graph. *Test: Same benchmark, Western-trained vs SA-tested, with architectural ablation removing the causal layer.*

P6: Ethical constraints improve prediction, not just compliance. Adding differentiable ethical constraints (LNN hard rules) to a consumer-decision model will improve out-of-distribution prediction accuracy (not just ethical compliance) by regularising the model against exploitative correlations that do not generalise. *Test: Ablation study: NSDM-F with vs without LNN constraint layer on held-out OOD test sets with demographic shift.*

P7: Imprecise consumer inference strengthens social conformity. Consumers under cognitive load, time pressure, or information overload will show stronger social-proof effects (brand conformity, review-score anchoring) than consumers with full deliberative capacity, because cognitive constraints reduce effective sensory precision, increasing the weight of social priors—replicating the Heins et al. finding that imprecise sensing enhances collective cohesion. *Test: Within-subject experiment: consumer choice under time pressure vs deliberative conditions, measuring social-proof effect size.*

7. Open Questions and Limitations

We are explicit about what this paper does not prove. The Variational Decision Hypothesis is a conjecture. The mathematical correspondences in Section 2 are structural observations, not derivations from first principles—we show that five frameworks share the same functional form, not that they must. The extension to consumer markets in Section 4 is a theoretical proposal with testable predictions but no empirical validation yet. The efficiency thesis in Section 5 depends on the claim that decision domains have low intrinsic dimensionality, which is true for bounded domains but may not hold for open-ended creative tasks.

Critical open questions: (1) Does the JEPa-to-free-energy correspondence hold *exactly* or only approximately? The contrastive/distillation losses in VICReg are not identical to the KL divergence—they are surrogates. Under what conditions does the approximation break? (2) Can the flocking-to-market mapping be made quantitatively precise, or is it limited to qualitative prediction? What are the right units for ‘distance in preference space’? (3) How does the architecture scale with the number of causal variables? Pearl’s do-calculus is computationally hard in the worst case. (4) Is the 5–10% informed-agent

threshold for trend adoption a robust prediction or an artefact of the Heins-Friston-Couzin model's specific parameterisation?

7.1 Relation to Quantum Cognition

A sixth framework demands explicit address: the quantum-cognition programme of Busemeyer and Bruza (2012) and the quantum decision theory of Yukalov and Sornette (2009). These model human choice using the mathematics of quantum probability—noncommutative judgements, interference between prospects, and non-additive probabilities—and successfully reproduce the same expected-utility paradoxes (Allais, Ellsberg, conjunction, order effects) that motivate our active-inference account. A reasonable reviewer will ask why we do not simply add quantum cognition as a sixth instantiation of the Variational Decision Hypothesis.

Our position is that quantum cognition and the free-energy account are *complementary descriptions at different levels*, not competitors. Quantum-probability models are *phenomenological*: they fit the statistical structure of choice (interference, contextuality) with remarkable economy but do not commit to a generative mechanism. The free-energy account is *mechanistic*: it specifies what the agent is computing (surprise minimisation over a generative model) and why. The two meet at a precise point—the non-commutativity of sequential judgements in quantum cognition corresponds, in our framework, to the order-dependence of belief updates when precision is reallocated between successive observations. Where the quantum formalism writes interference terms, active inference writes precision-weighted prediction errors that have not yet equilibrated. We conjecture that the quantum-probability formalism is the equilibrium statistics of a free-energy process observed before convergence—a testable claim, since it predicts that interference effects should diminish as deliberation time increases and beliefs equilibrate.

This connection is not idle. Yukalov and Sornette's broader programme links quantum decision theory to the prediction of critical transitions in collective systems—Sornette's log-periodic power law (LPPL) detection of pre-crash regimes in financial markets. That work is the rigorous, empirically validated antecedent of our Prediction P2 (market panics

as precision-modulated phase transitions). We adopt LPPL critical-point detection as the methodological backbone for the market-fragility application, treating it as the directional-agnostic, regime-detection complement to the flocking model rather than reinventing crash-precursor mathematics.

We also acknowledge an epistemological limitation. This paper is written by a practitioner-researcher at a one-person AI company in Johannesburg, not by a team at a frontier lab with access to thousand-GPU clusters. The theoretical contributions stand or fall on their mathematical merit, not institutional affiliation. But the empirical validation programme described in Section 6 will require collaborators, data, and compute that exceed a single operator's resources. This paper is published as a living document precisely because it is an invitation to collaboration, not a finished edifice.

8. Conclusion: Intelligence is One Thing

Euler's insight about the bridges of Königsberg was that the problem was not about bridges at all. It was about the topology of networks—a structure that existed independently of the physical bridges, rivers, and landmasses of that Prussian city. Our claim is analogous. The five research programmes surveyed here are not building toward five different theories of intelligence. They are building toward one theory, expressed in five different notations, constrained by five different disciplinary traditions. The variational free energy functional is the topology underlying all of them.

If this is correct, the implications extend beyond academic interest. It means that the dominant approach to AI—scaling monolithic neural networks to unprecedented size—is building increasingly powerful approximations to one component (Layer 1: the world model) while ignoring four others. It means that the regulatory challenges of AI transparency and accountability are not unsolvable engineering problems but architectural deficiencies: a system without Layer 2 (causal-symbolic reasoning) cannot explain itself because it has nothing to explain *with*. It means that the emerging markets systematically excluded from frontier AI deployment are not permanently behind but

potentially ahead—because their infrastructure constraints select for exactly the architecture the mathematics recommends.

And it means that the next generation of decision-intelligence systems will not be built by making GPT-5 bigger. They will be built by making intelligent systems *whole*.

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MATHEMATICAL APPENDICES

Appendix A. The JEPA ↔ Free Energy Correspondence

This appendix formalises the claim that LeCun’s Joint Embedding Predictive Architecture (JEPA) objective and Friston’s variational free energy are equivalent under specific conditions. We proceed in three steps: state the JEPA objective, state the free energy functional, and derive the mapping.

A.1 The JEPA Objective

In JEPA (LeCun, 2022; Assran et al., 2023, 2025), the system consists of: (1) a context encoder f_θ mapping context input x to representation $s_x = f_\theta(x)$; (2) a target encoder f_ξ mapping target input y to $s_y = f_\xi(y)$; and (3) a predictor g_φ mapping context to predicted target: $\hat{s}_y = g_\varphi(s_x)$. The objective minimises:

$$L_{JEPA} = D(s_y, \hat{s}_y) + R(s_x, s_y)$$

where D is a distance function (L2 or cosine) and R prevents representation collapse. In VICReg (Bardes et al., 2022):

$$R_{VICReg} = \lambda \cdot v(s) + \mu \cdot Cov(s) + \nu \cdot \|s\|$$

where $v(s)$ enforces per-dimension variance above a threshold, $Cov(s)$ decorrelates dimensions, and $\|s\|$ is the invariance term. In V-JEPA 2, the target encoder is an exponential moving average: $\xi \leftarrow \tau\xi + (1-\tau)\theta$, providing implicit regularisation.

A.2 The Variational Free Energy Functional

Friston’s variational free energy for an agent with observations o , hidden states θ , approximate posterior $q(\theta)$, and generative model $p(o, \theta)$ is:

$$F[q] = E_q[\ln q(\theta) - \ln p(o, \theta)]$$

Decomposition 1 (Energy – Entropy): $F = E_q[-\ln p(o, \theta)] - H[q(\theta)]$

Decomposition 2 (KL + Surprise): $F = D_{KL}[q(\theta) \parallel p(\theta | o)] + [-\ln p(o)]$. Since $D_{KL} \geq 0$, free energy is an upper bound on surprise. The bound is tight when $q(\theta) = p(\theta | o)$.

A.3 The Correspondence

Proposition A.1. Under Gaussian assumptions on $q(\theta)$ and $p(o | \theta)$, the JEPA L2 prediction loss $\|s_y - g(s_x)\|^2$ is proportional to the energy term of variational free energy.

Proof. If $p(o | \theta) = N(o; f(\theta), \sigma^2 I)$, then:

$$-\ln p(o | \theta) = (1/2\sigma^2)\|o - f(\theta)\|^2 + \text{const}$$

Identifying $o \rightarrow s_y$ and $f(\theta) \rightarrow g(s_x)$:

$$E_q[-\ln p(o | \theta)] = (1/2\sigma^2) E_q[\|s_y - g(s_x)\|^2] + \text{const}$$

The energy term of free energy is the JEPA prediction loss scaled by inverse precision $1/2\sigma^2$.

□

Proposition A.2. The VICReg variance regulariser lower-bounds the entropy of the approximate posterior $H[q(\theta)]$.

Proof. For a d -dimensional distribution, entropy is maximised by the Gaussian with the same covariance. For a factored Gaussian, $H[q] = (d/2)\ln(2\pi e) + (1/2)\sum \ln \sigma_i^2$. VICReg enforces $\sigma_i^2 \geq \gamma$, yielding $H[q] \geq (d/2)\ln(2\pi e) + (d/2)\ln \gamma$. The covariance decorrelation pushes toward a factored Gaussian (maximising entropy for given variances). Together, the VICReg regulariser maximises a lower bound on $H[q(\theta)]$, serving the same role as the entropy term $-H[q]$ in F . □

Corollary A.3. Under Gaussian assumptions:

$$L_{JEPA} = (1/2\sigma^2)\|s_y - g(s_x)\|^2 + R_{VICReg}(s) \geq F[q] + \text{const}$$

JEPA over-estimates free energy, which is conservative: the system believes the world is more surprising than it is, erring toward exploration.

A.4 Where the Correspondence Breaks

Three disanalogies: (1) JEPA's EMA target encoder has no direct variational inference analogue—it resembles a target network in DQN or slowly evolving prior in empirical Bayes. (2) The Gaussian assumption may not hold in learned representation spaces with complex geometry. (3) Free energy is defined over probability distributions; JEPA operates

on point embeddings. The connection is exact only when JEPA is interpreted as a Laplace approximation (MAP estimate + Hessian-based uncertainty) to the full variational objective—standard in computational neuroscience (Friston et al., 2007) but introducing approximation error proportional to posterior non-Gaussianity.

Appendix B. The Drift-Diffusion Model as Active Inference

This appendix demonstrates that the DDM is a special case of active inference where the agent's only actions are 'continue sampling' and 'commit.'

B.1 Sequential Bayesian Reformulation

Consider binary hypothesis testing where observations arrive sequentially. Under H_1 : $o_t \sim N(\mu, \sigma_o^2)$; under H_0 : $o_t \sim N(-\mu, \sigma_o^2)$. The log-posterior ratio after n observations:

$$L_n = L_0 + \sum_{t=1}^n (\mu/\sigma_o^2) \cdot o_t$$

In continuous time:

$$dL = (\mu^2/\sigma_o^2) dt + (\mu/\sigma_o) dW$$

This is a drift-diffusion process with $v = \mu^2/\sigma_o^2$ and $\sigma = \mu/\sigma_o$. The DDM is *exactly* sequential Bayesian inference under Gaussian likelihoods.

B.2 Active Inference Derivation

The agent has two policies: π_s (sample) and π_c (commit). Expected free energy of sampling:

$$G(\pi_s) = -E_q[\text{info gain from } o_{t+1}] + \text{cost}_{\text{sample}}$$

Expected free energy of committing:

$$G(\pi_c) = -\ln q(\theta^*) + \text{cost}_{\text{error}}$$

The commit condition $G(\pi_c) < G(\pi_s)$ is equivalent to the posterior exceeding a threshold—precisely the DDM boundary a .

Theorem B.1. The DDM with drift v and boundary a is equivalent to active inference with Gaussian generative model, policies $\{\pi_s, \pi_c\}$, and expected free energy with pragmatic precision γ , under:

$$v = \partial F / \partial \mu = \text{precision} \times \text{prediction error}$$

$$a = f(\gamma, \text{cost}_{\text{error}}, \text{cost}_{\text{sample}})$$

$$z/a = \text{prior bias} / \text{boundary} = \text{normalised prior log-odds}$$

The drift rate is the gradient of free energy with respect to beliefs. Evidence accumulation is gradient descent on free energy. The boundary is the precision-weighted trade-off between error and sampling costs. The DDM and active inference are not analogous—they are identical.

B.3 Extension to the Neural DDM

In NSDM-F, the drift rate is parameterised as a learned function of the world model’s latent state:

$$v(s_t; \theta_v) = W_v \cdot [s_t^{\{det\}} \oplus s_t^{\{stoch\}}] + b_v$$

where $s_t^{\{det\}}$ and $s_t^{\{stoch\}}$ are from Layer 1’s RSSM. Under the active inference interpretation, prediction error is computed in the world model’s representation space—the JEPA principle. The N-DDM drift rate is the gradient of a latent-space free energy functional, where the latent space is structured by learned dynamics.

Appendix C. LNN Constraint Satisfaction as Variational Inference on Factor Graphs

IBM's LNNs maintain truth-value bounds $[l_i, u_i] \in [0,1]^2$ for each proposition, with inference as bound-tightening message passing. We show this is variational inference on a factor graph where variable nodes are propositions and factor nodes are logical constraints.

C.1 Factor Graph Formulation

Define factor graph $G = (X, C, E)$ with proposition variables X , constraint factors C , and edges E . Each constraint c_j encodes a logical formula as a real-valued function using Fréchet bounds:

$$\begin{aligned} \text{AND: } f_{\wedge}(a,b) &= \max(0, a+b-1) & \text{OR: } f_{\vee}(a,b) &= \min(1, a+b) \\ \text{NOT: } f_{\neg}(a) &= 1-a & \text{IMPLIES: } f_{\rightarrow}(a,b) &= \min(1, 1-a+b) \end{aligned}$$

The joint distribution:

$$p(x_1, \dots, x_n) \propto \prod_j f_j(x_{\{j_1\}}, \dots, x_{\{j_k\}})^{w_j}$$

This is a Markov random field with logical-semantic potential functions.

C.2 LNN Inference as Belief Propagation

The LNN's upward-downward pass is structurally isomorphic to loopy belief propagation on this factor graph, with interval-valued messages. Convergence corresponds to a fixed point of the Bethe free energy:

$$F_{\text{Bethe}} = \sum_j E_j \{b_j\} [-\ln f_j] + \sum_i (d_i - 1) H[b_i]$$

where b_j are local beliefs at factor j , b_i are marginal beliefs at variable i , and d_i is the degree of variable i . The Bethe free energy approximates the variational free energy of the full joint. Therefore, **LNN inference is an interval-arithmetic variant of variational inference on a factor graph defined by the domain's logical structure**. For the NSDM-F ontology of ~50 rules, the factor graph is typically sparse and nearly tree-structured, ensuring good convergence.

Appendix D. The Information Bottleneck and Rate-Distortion Theory

Layer 5's conditional mutual information term $I(S; A / O)$ regularises policy complexity. This appendix connects it to the information bottleneck (Tishby et al., 2000) and rate-distortion theory (Shannon, 1959).

D.1 The Policy Bottleneck

The information bottleneck for a compressed representation T of X predictive of Y :

$$L_{IB} = I(X; T) - \beta \cdot I(T; Y)$$

For a decision agent with states S , actions A , rewards R :

$$L_{policy} = I(S; A) - \beta \cdot I(A; R)$$

Use the minimum-complexity policy that maintains maximum predictive power over outcomes. This connects to maximum-entropy RL (Haarnoja et al., 2018):

$$J(\pi) = E[\sum_t r_t - \alpha \ln \pi(a_t | s_t)]$$

The entropy regulariser $-\alpha \ln \pi$ upper-bounds $I(S; A)$ and prevents overfitting to irrelevant state distinctions.

D.2 Why This Favours Small Models

Rate-distortion theory proves that for decision domains with intrinsic dimensionality k embedded in d -dimensional observation space ($k \ll d$), the rate-distortion function scales with k , not d . A consumer choosing among n products based on k latent dimensions needs only $O(k \cdot \log n)$ bits of policy complexity—regardless of stimulus dimensionality. A 14B SLM with causal structure can saturate this bound, making further scaling wasteful. The SLM's capacity matches the decision's intrinsic dimensionality, not the stimulus surface dimensionality.

Appendix E. Derivation of the Cross-Cultural Precision Hypothesis

This appendix derives Prediction P1 and the Cross-Cultural Precision Hypothesis from the Heins-Friston-Couzin (2024) active inference model.

E.1 The HFC Agent Model

Each agent i maintains beliefs $\mu_i = \{\mu_i^1, \dots, \mu_i^L\}$ about expected neighbour distances in L visual sectors. Generative model:

$$o_i^l \sim N(\mu_i^l, \pi_s^{-1}) \quad \mu_i^l \sim N(\mu_0, \pi_p^{-1})$$

where π_s is sensory precision and π_p is prior precision. Belief dynamics follow:

$$d\mu_i^l/dt = \pi_s(o_i^l - \mu_i^l) + \pi_p(\mu_0 - \mu_i^l)$$

Steady-state:

$$\mu_i^{l*} = (\pi_s \cdot o_i^l + \pi_p \cdot \mu_0) / (\pi_s + \pi_p)$$

When $\pi_s \gg \pi_p$: beliefs track observations (agent trusts its eyes). When $\pi_p \gg \pi_s$: beliefs stay near prior (agent trusts expectations). **Low sensory precision produces stronger collective cohesion** (Heins et al., 2024).

E.2 Consumer Market Translation

HFC Parameter	Consumer Interpretation	Cultural Dimension
π_s (sensory precision)	Confidence in own product evaluation	Higher in individualist cultures (IDV > 70)
π_p (prior precision)	Strength of social expectations	Higher in collectivist cultures (IDV < 30)
μ_0 (prior mean)	Default choice pattern	Culturally determined per category
o_i^l (observation)	Observed peer choices in sector l	Social media, word-of-mouth

Table E.1. HFC-to-consumer parameter mapping.

E.3 The Prediction

Define the social conformity index C_i as the correlation between consumer i 's choices and their peers' average. From the steady-state equation:

$$C_i = \text{Corr}(\text{choice}_i, \text{mean}(\text{choices}_{\{\text{peers}\}})) \propto \pi_p / (\pi_s + \pi_p)$$

Identifying Hofstede's individualism score IDV with the precision ratio π_s/π_p :

$$C_i \propto 1 / (1 + IDV_{\text{culture}} / k)$$

where k maps Hofstede's 0–100 scale to precision ratios. **Brand-switching rates are an inverse function of individualism index, mediated by the precision ratio in a variational free energy model of social decision-making.**

For South Africa: multiple cultural clusters (Nguni $IDV \approx 20$, European-descent urban $IDV \approx 65$, Sotho-Tswana and Indian-descent intermediate) allow within-country testing that controls for macro-economic conditions.

E.4 Cognitive Load as Precision Modulation

If cognitive load reduces effective sensory precision π_s , the social prior π_p gains relative weight. Prediction P7: **the same consumer under cognitive load shows stronger social-proof effects than under deliberative conditions**—replicating HFC's finding that imprecise sensing enhances collective cohesion. Test: product-choice scenarios with/without social-proof cues, under deliberative and time-pressured conditions. Predicted interaction: social-proof influence $\times$ cognitive-load, magnified under load.

Colophon

This is a living document. It will be updated as (a) empirical results from the prediction programme become available, (b) new results from the five research programmes either strengthen or weaken the Variational Decision Hypothesis, and (c) the South African AI regulatory landscape crystallises. Version history will be maintained at the MetaForgeAI research repository.

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